

Barath R

Electronics & Communication Engineering Student | Embedded Systems & IoT

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Professional Summary

Third-year Electronics and Communication Engineering student with hands-on experience across embedded firmware, bare-metal systems programming, and IoT hardware design. Comfortable working close to the metal – from register-level RISC-V trap handlers to production-style sensor and communication pipelines – and building the cloud/dashboard layers that sit on top of them. Seeking an embedded systems internship/role to contribute to real-time and connected hardware products.

Education

B.E. Electronics and Communication Engineering 2024 – 2028 (Expected)
Rajalakshmi Institute of Technology, Chennai CGPA: 8.65/10

Currently in 3rd year

Higher Secondary Certificate (HSC) – Class XII 2023 – 2024
Montfort Matriculation Higher Secondary School, Chennai Percentage: 95%

Tamil Nadu State Board

Secondary School Leaving Certificate (SSLC) – Class X 2021 – 2022
Montfort Matriculation Higher Secondary School, Chennai Percentage: 94%

Tamil Nadu State Board

Technical Skills

- **Hardware & Development Boards:** ESP32, ESP32-C6, ESP32-CAM, Raspberry Pi 5, Arduino UNO R3, Altera DE2 Development Kit
- **PCB Design & EDA Tools:** PCB/PCD Design, KiCad, Proteus, Schematic Capture, 2/4-Layer PCB Layout, Routing, DRC/ERC, Gerber & BOM Generation, Soldering & Board Bring-up
- **Communication Protocols:** MQTT, LoRaWAN, I2C, SPI, UART, HTTP/REST, WebSockets
- **Programming Languages:** C, Python, C++, Java, SQL
- **Cloud & Dashboards:** HiveMQ, ThingSpeak, Firebase, BlynkIoT, Google Cloud
- **Tools:** Git, VS Code, PlatformIO, ESP-IDF (v6), Arduino IDE, MATLAB, Scilab, CST Studio, Keysight ADS
- **Android/OS:** ADB, custom OS/ROM flashing

Featured Projects

Bare-Metal RISC-V Trap Handler & Context Switcher *ESP32-C6*

Tech Stack: RISC-V Assembly, direct MMIO, no RTOS

- Implemented a trap handler and timer-based context switcher entirely in assembly with no FreeRTOS or vendor libraries
- Worked directly with memory-mapped I/O registers for interrupt handling and task switching

POWER GUARD – ESP32-C6 Smart Power Monitor *ESP32-C6 / Embedded Firmware*

Tech Stack: C++, ESP-IDF, FreeRTOS, I2C (SH1106 OLED), ThingSpeak IoT, ADC, Relay Control

- Developed embedded firmware on ESP32-C6 for real-time AC voltage/current monitoring and GPIO10 overload cut-off protection
- Implemented modular SH1106 OLED driver over I2C with 10-phase diagnostics boot sequence and ThingSpeak cloud telemetry

Dual-Axis Solar Tracker *Embedded System*

Tech Stack: Servo Motors, LDR Sensors, Microcontroller Control

- Built a dual-axis solar panel tracking system to maximize sunlight incidence using closed-loop sensor feedback
- Implemented motor control logic for continuous azimuth and elevation tracking

Digital Twin for Industrial Monitoring *5G-Integrated IoT System*

Tech Stack: Sensor Fusion, 5G Connectivity, Real-Time Dashboard

- Developing a digital twin platform for industrial equipment monitoring with 5G-integrated data pipelines
- Designing real-time sensor fusion and visualization for predictive maintenance use cases

Conformal Helmet-Integrated Antenna Array *RF / Antenna Design*

Tech Stack: Microstrip Patch Array, Dual-Band UHF/L-Band

- Designing a flexible microstrip patch antenna array conformal to a helmet surface, replacing bulky whip antennas
- Targeting dual-band UHF and L-band support with an RF shielding layer and ruggedized coax interface to existing radio/camera systems

Technical Experience

- Conceptualized, developed, and iterated on multiple embedded and IoT systems end-to-end, from hardware bring-up to firmware and dashboards
- Active organizer with IETE Students' Forum (ISF), coordinating technical hackathons and events